

Clinical Image Analysis and Enhancement Workbench

Medical Image Processing & Computer Vision
Group Project

9 teams (4 members) + 2 teams (5 members)
20 Marks

General Rules:

GUI Framework: Desktop App → Python (PyQt/CustomTkinter) or MATLAB (App Designer).

No Built-in Functions: For example, functions for spatial filtering, interpolation, convolution, and histogram equalization must be implemented entirely from scratch.

Permitted Built-ins: Complex mathematical transformations like `fft2`, `ifft2`, `fftshift`, and `ifftshift` are permitted to save execution time, but the filter masks themselves must be generated manually.

Error Handling: The application must not crash. It should gracefully handle corrupted images, unsupported formats, or invalid user inputs.

Code Plagiarism & Originality: All submitted code will be run through plagiarism detection software. While the mathematical logic behind these algorithms is universal, your team's specific implementation, software architecture, and vectorization strategies must be completely your own. Teams caught sharing code, copying from past cohorts, or submitting blatantly copy-pasted AI-generated scripts will face penalties.

Mandatory Technical Documentation: Along with the final code, every team must submit a formal technical document. This document **MUST** include a "Contribution Matrix" outlining the exact code modules, GUI components, and testing phases each individual team member was responsible for building.

Mandatory In-Person Discussion (The Defense): The final grade is heavily dependent on an in-person, live defense of your codebase. Submitting the code is only half the requirement. During the discussion:

- **Code Discussions:** Each team member will explain specific blocks of code they were responsible for building. However, the whole team is responsible for understanding the entire workbench.
- **Live Modifications:** Be prepared to modify your code on the spot. You may be asked to instantly change an edge-handling strategy, alter a boundary condition, or optimize a nested loop.

Phase 1: Spatial Domain Operations & Core Architecture (Week 13) 10 MARKS

Objective: Establish the GUI, handle image I/O, and implement foundational spatial enhancements.

1. Multi-Format Image I/O and Viewer

- Create a tabbed or multi-window GUI that loads and displays DICOM, JPEG, and BMP images .
- Extract and display relevant metadata (Width, Height, Bit depth, and DICOM-specific tags like Modality, Patient Name, Age, and Body Part Examined) .
- **Save Functionality:** Implement a button to export and save the currently displayed (processed) image to the local drive.
- **Interactive Zooming:** Implement Zoom In and Zoom Out controls. **Crucially, this must be powered by the team's custom Interpolation algorithms.** Built-in zoom libraries are strictly forbidden. The user must be able to select whether the zoom utilizes Nearest-Neighbor or Linear interpolation (written from scratch).

2. Custom Spatial Filtering Engine

- Implement a 2D convolution function from scratch.
- Allow the user to select a kernel size (3 x 3, 5 x 5, etc.) and apply:
 - **Smoothing:** Average and Gaussian filters (user inputs the variance for Gaussian).
 - **Edge Detection:** Sobel or Prewitt operators (displaying horizontal, vertical, and combined magnitude edges).
 - **Non-linear Filtering:** Median filter from scratch to handle noise.

3. Advanced Histogram Processing

- Instead of standard global equalization, implement **Local Histogram Equalization**. The user inputs a block size (e.g., 8 x 8 or 16 x 16), and the algorithm equalizes local regions to enhance local contrast, specifically useful for medical images with uneven illumination.

4. The "Sequential Enhancement" Pipeline Mode

- Implement a state-management system within the GUI that allows users to stack operations iteratively.
- Users must be able to apply an operation (e.g., a Median filter to remove salt-and-pepper noise) and then apply a subsequent operation *on top of that exact result* (e.g., Local Equalization to fix contrast), viewing the cumulative clinical enhancement.

- Include an "Undo Last Step" and a "Reset to Original" button to manage this pipeline.

Extra Deliverable for 5-Member Teams (Phase 1):

- **Geometric Transformations:** Implement image rotation and shearing from scratch. Since the coordinates will land on non-integer pixels, the team must use the Bilinear Interpolation method (implemented from scratch) to calculate the new pixel values .

Phase 2: Frequency Domain, Noise Analysis, and Segmentation (Week 14) 10 MARKS

Objective: Transition to the frequency domain to handle complex, periodic clinical noise, and introduce interactive statistical analysis of specific tissue regions.

1. Interactive Notch Filtering (Periodic Noise Removal)

- **The Clinical Problem:** Medical scans (especially older digitized films or equipment with electronic interference) often suffer from periodic noise, which appears as a repeating pattern over the image and as distinct, bright "stars" in the Fourier spectrum.
- **The Implementation:** Transform the image to the Fourier domain and display the shifted magnitude spectrum (using log scaling for visibility) .
 - **Interactive GUI:** Allow the user to click directly on the bright noise spikes within the displayed magnitude spectrum.
 - **The Mathematical Engine:** Capture the clicked (u, v) pixel coordinates and automatically generate a **Notch Reject filter** (user chooses between Ideal, Butterworth, or Gaussian shapes).
 - *Crucial Constraint:* To ensure the output remains a real image (no complex numbers), the algorithm must automatically mirror the user's click and place a conjugate notch in the symmetric opposite quadrant of the spectrum.
 - Multiply the generated mask with the Fourier transform and apply the inverse transform to display the cleaned image .

2. ROI Statistical Analysis & Noise Modeling

- Implement a tool to inject synthetic noise (Gaussian and Uniform) into an image from scratch .
- **Interactive ROI Tool:** Allow the user to draw a rectangular Region of Interest (ROI) directly onto the image.
- **Local Statistics:** Isolate the pixels within that ROI, compute its local histogram from scratch, and display the histogram alongside the exact mean and variance of that specific tissue region .

Extra Deliverable for 5-Member Teams (Phase 2):

- **Frequency Domain Template Matching (Cross-Correlation):** The team must allow the user to crop a small template from the image (e.g., a specific cell or bone fragment). Using the Fourier Domain, implement cross-correlation to find and draw a bounding box around the template's location within the original image.

Bonus Task: Clinical Morphological Engine

4 MARKS

(No Partial Grading / Complete Mandatory Reqs. First)

Objective: Implement fundamental binary set-theory operations to clean, segment, and enhance thresholded medical images.

- **The Clinical Problem:** After a medical image is segmented or thresholded (turned into black and white), the resulting mask is rarely perfect. It often contains salt-and-pepper noise, ragged edges, or gaps inside structures like blood vessels. Morphological operations correct these spatial defects.
- **The Task:** Implement the four core binary morphological operations entirely from scratch. You are forbidden from using morphological libraries (e.g., `cv2.erode`, `skimage.morphology`, `bwmorph`).
- **The Algorithm Requirements:**
 1. **Thresholding:** First, implement a simple global thresholding slider to convert a loaded grayscale image into a binary mask.
 2. **Core Operations:** Write the 2D spatial logic for **Erosion** and **Dilation**.
 3. **Compound Operations:** Using your custom Erosion and Dilation functions as building blocks, build **Opening** (Erosion followed by Dilation) to remove small noise, and **Closing** (Dilation followed by Erosion) to fill in gaps.
- **GUI Integration:** The user must be able to binarize the image.
 - The user must be able to select the size (e.g., 3x3, 5x5) and shape of the **Structuring Element (SE)**. At minimum, support a Square and a Cross SE.
 - Provide buttons to apply Erosion, Dilation, Opening, or Closing, and display the result immediately in the pipeline viewer.
- **Extra Credit (Optional 5-Member Team Scope):** Implement a morphological **Boundary Extraction** algorithm (subtracting an eroded image from the original binary image) to outline anatomical structures.

Team Number:

Name	ID	Contribution “Detailed”

Project Defense Evaluation 35%:

To be filled by TA

Name	Evaluation(Phase 1 / 2)
Student 1	
Student 2	
Student 3	
Student 4	
Student 5	

Deadlines

Phase 1 submission through Google Classroom by 7:59 AM – MAY 5th 2026

Phase 1 discussions Tuesday – MAY 5th & Thursday – MAY 7th

Phase 2 submission through Google Classroom by 11:59 PM MAY 9th 2026

Phase 2 discussions – TBC

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